ENEE452 Project : myDACQ

ST's B-L475E-IOT01A board provides numerous methods for data-acquisition (DACQ) and a variety of methods for communicating the results. Your recent homework assignments have addressed different parts of such systems. The myDACQ project is the integration of those parts, in two steps:

myDACQ Project phase 1 – standalone demo of your Homework 08 device and display connection

myDACQ Project Phase 2 – connecting device and display through a wifi-connected MQTT server

Phase 1) communications between your myDACQ and CubeIDE's serial console menu.
Data generated by your device will be messagepack'ed and sent to the console for unpacking and display. Commands from your console menu will be messagepack'ed and sent to your device for configuration. Messages are to be exchanged between device and console using Homework 07's 'msgPost()' methods. For messagepack use 'CWPack' (https://github.com/clwi/CWPack/blob/master/README.md).

Phase 2) remoting myDACQ – routing messagepack'ed data through a wifi-connected MQTT server
Messages between your myDACQ device and its display will be instead posted to an outgoing MQTT client/wifi radio's input queue to be received by the PC/MQTT server. The PC/MQTT server will immediately echo them back to an incoming MQTT client/wifi radio to be posted to the appropriate display or device input queue. (For MQTT we have recommended https://github.com/kokke/tiny-MQTT-c)

(Note: implementing "outgoing MQTT client/wifi radio", "PC/MQTT server", and "incoming MQTT client/wifi radio" will require integration of Homework 10's Problem A and Problem B solutions. We will be publishing the solutions received from the class after a few day's grace to allow Homework 10 late submissions to be counted toward the Homework 10 grade. To get started on the inward-facing parts of Phase 2 you should look at connecting the outgoing and incoming MQTT client to the various device and display message queues involved.)